Brainstorm & Ideation Phase

Date	28-10-2026
Team ID	SWTID-2026-7875
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Brainstorm & Idea Prioritization Template:

Brainstorming Categories:

Security & Script Robustness

- **Dynamic Role Checking:** Refactor hardcoded bb1 checks to query user branch via `sys_user` rather than relying on custom static roles.
- **Script Edge Case Handling:** Gracefully handle users lacking explicit branch assignments without granting default open access.
- **Data-Level Row Security:** Implement `before query` Business Rules alongside Read ACLs to eliminate list security constraint warnings.

Scalability & Architecture

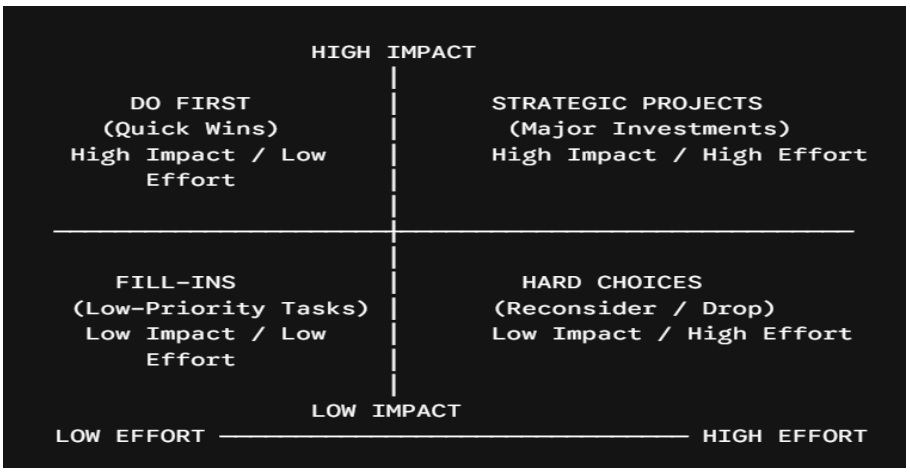
- **Role Hierarchy Consolidation:** Group granular roles (bb1–bb4) into structured inheritance models (e.g., Read, Write, Manager).
- **Multi-Branch Multi-Tenancy:** Expand single-branch logic into a dynamic matrix supporting ECE, CSE, and EEE branches seamlessly.

UX & Field Security

- **Field-Level Security:** Add targeted field ACLs on PII elements (e.g., Email, Phone Number) for granular record protection.
- **UI Policy Synchronization:** Align form UI policies with underlying ACL permissions to cleanly hide or disable read-only fields.

IDEA PRIORITIZATION MATRIX (IMPACT VS. EFFORT)

Categorize ideas based on their evaluation scores into the four execution quadrants:



Idea Prioritization Framework:

IDEA DESCRIPTION	CATEGORY	IMPACT	EFFORT	PRIORITY TIER
Dynamic `sys_user` Query Replace hardcoded script conditions with dynamic user queries.	Security	High	Low	P1 (QUICK WIN)
Role Inheritance Design Consolidate bb1–bb4 into structured parent roles.	Scalability	High	Low	P1 (QUICK WIN)
`Before` Query Business Rules Enforce data-level row restrictions before display.	Security / UX	High	Medium	P2 (MAJOR)
Extend Policy to ECE & CSE Expand single-branch logic to multi-tenancy model.	Scalability	Medium	Low	P2 (MAJOR)
Field-Level PII Restrictions Protect sensitive fields (Email, Phone) via field ACLs.	Security	Medium	Medium	P3 (FILL-IN)
Automated ATF Test Suite Build Automated Test Framework scripts for ACL coverage.	QA / Testing	Low	High	P4 (RECONSIDER)